

# Taming Multiparticle Entanglement

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## Taming Multiparticle Entanglement

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### Keywords:

- **Biseparable States**
- Efficient criterion for **genuine multipartite entanglement (GME)** via **PPT mixtures**
- **Semidefinite programming (SDP)** formulation for practical detection
- **Fully decomposable entanglement witnesses**
- **Entanglement Quantification**

## 1. Motivation

- Separability & entanglement
- Examples of multipartite states
- Challenges in detection

## 2. Definitions

- Biseparable states
- Genuine multipartite entanglement (GME)

## 3. Main Idea

- PPT mixtures
- Fully decomposable witnesses
- SDP formulation

## 4. Results

- White noise tolerance
- Entanglement monotone

## 5. Summary

# Motivation

**The Challenge:** How do we efficiently verify that a state has *genuine* multipartite entanglement that cannot be reduced to separable states?

## Separability and Entanglement

A state  $\rho$  is **separable** if:

$$\begin{aligned} |\psi\rangle &= |\psi_A\rangle \otimes |\psi_B\rangle \\ \rho &= \sum_i p_i \rho_i^A \otimes \rho_i^B \end{aligned}$$

where  $\sum_i p_i = 1$  and  $p_i > 0$ .

**Problem:** Directly determining separability is **NP-hard**!

- How should we split the system into subsystems?
- It is a costly operation!

## [Motivation] Verification of Entangled States

For pure states, SVD is the most effective way to quantify entanglement. For example, consider the two-qubit Bell state (though this is a trivial example).

**Step 1.** Write the state.  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + 0|01\rangle + 0|10\rangle + \frac{1}{\sqrt{2}}|11\rangle = \sum_{i,j} c_{ij}|i\rangle_A|j\rangle_B$

**Step 2.** Form Coefficient Matrix.

$$C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

**Step 3.** Find SVD of  $C$ , that is,  $C = U\Sigma V^\dagger$ .

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

**Step 4.** Two Schmidt coefficients  $\Rightarrow$  entangled state. Try the other Bell states!

## [Motivation] Challenges for Entanglement Detection: Pure States

Exponential cost is inevitable. For an  $m \times n$  matrix:

| Resource | Complexity                                                              |
|----------|-------------------------------------------------------------------------|
| Time     | $\mathcal{O}(\min(mn^2, m^2n)) \simeq \mathcal{O}(mn \cdot \min(m, n))$ |
| Memory   | $\mathcal{O}(mn)$ for matrix + $\mathcal{O}(m^2 + n^2)$ for $U, V$      |

For  $N$  qubits with bipartition  $k|(N - k)$ , the matrix size becomes  $2^k \times 2^{N-k}$ . Things get worse if we must try every possible bipartition, of which there are  $2^{N-1}$ .

## [Motivation] Challenges for Entanglement Detection: Mixed States

Even worse, for mixed states there is no "canonical form" like the Schmidt decomposition. The same  $\rho$  can be written as:

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k| = \sum_j q_j |\phi_j\rangle\langle\phi_j|$$

with completely different ensembles! One might use only product states (separable), another might not (entangled representation), while the **physical state is the same**.

# [Definition] Biseparable States, Genuine Multipartite Entanglement

A state is separable with respect to the bipartition  $M|\bar{M}$  if:

$$\rho_{M|\bar{M}}^{\text{sep}} = \sum_i p_i \rho_M^{(i)} \otimes \rho_{\bar{M}}^{(i)}$$

A state is **biseparable** if it can be written as a mixture:

$$\rho^{\text{bs}} = \sum_i p_i \rho_{M_i|\bar{M}_i}^{\text{sep}}$$

where  $\sum_i p_i = 1$ . Note that a non-trivial form (with at least two  $p_i \neq 0$ ) must be a mixed state.

## Genuine Multipartite Entanglement (GME)

A state is **genuine multipartite entangled** if and only if it is **NOT biseparable**.

$$\text{GME} = \text{NOT biseparable}$$



## [Example] Biseparable States

Consider a three-qubit system (labeled A, B, C). There are three possible bipartitions:  $A|BC$ ,  $B|CA$ , and  $C|AB$ , i.e.,  $M|\bar{M} \in \{A|BC, B|CA, C|AB\}$ . A biseparable state can be written as:

$$\rho^{\text{bs}} = p_{A|BC} \rho_{A|BC}^{\text{sep}} + p_{B|CA} \rho_{B|CA}^{\text{sep}} + p_{C|AB} \rho_{C|AB}^{\text{sep}}$$

## [Main Idea] Checking whether PPT mixture exists

It is well-known that separable states have Positive Partial Transpose (PPT). We denote such PPT states by  $\rho_{M|\bar{M}}^{\text{ppt}}$ . Thus, we ask whether a state can be written as

$$\rho^{\text{pmix}} = \sum_i p_i \rho_{M_i|\bar{M}_i}^{\text{ppt}}$$

We call states of this form **PPT mixtures**.

Clearly, any biseparable state is a PPT mixture, so proving that a state is *not* a PPT mixture implies **genuine multipartite entanglement**.

## [Recap] Positive Partial Transpose

For a bipartite state  $\rho_{AB}$  with matrix elements  $\rho_{ij,kl} = \langle i, j | \rho | k, l \rangle$ :

$$\rho_{ij,kl}^{T_A} = \rho_{kj,il}$$

A state has **Positive Partial Transpose (PPT)** if  $\rho^{T_A} \geq 0$ .

**Quick Example: Bell state**

$$\rho = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \rightarrow \rho^{T_A} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

**Eigenvalues of  $\rho^{T_A}$ :**  $\left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right\}$

The **negative eigenvalue** proves the Bell state is entangled!

# [Main Idea] Characterization via entanglement witnesses

## Definition

An observable  $W$  is an **entanglement witness** for GME if:

- $\text{Tr}(W\rho) \geq 0$  for all biseparable states  $\rho$
- $\text{Tr}(W\rho) < 0$  for some GME state  $\rho$

## Decomposable Witnesses

A witness is **decomposable** with respect to bipartition  $M|\bar{M}$  if:

$$W = P_M + Q_M^{T_M}$$

where  $P_M \geq 0$  and  $Q_M \geq 0$ .

A witness  $W$  is **fully decomposable** if for **all** bipartitions  $M$ :

$$W = P_M + Q_M^{T_M}, \quad P_M \geq 0, \quad Q_M \geq 0$$

# [Main Idea] Fully Decomposable Witnesses

## Theorem

Fully decomposable witnesses detect **exactly** the states that are **not PPT mixtures**.

If  $\rho$  is not a PPT mixture, then there exists a fully decomposable witness  $W$  that detects  $\rho$ .

$$\rho \notin \mathcal{S}_{\text{PPT-mix}} \implies \exists W \in \mathcal{W}_{\text{fd}} : \text{Tr}(W\rho) < 0$$

where

- $\mathcal{S}_{\text{PPT-mix}}$  = set of PPT mixtures
- $\mathcal{W}_{\text{fd}}$  = set of fully decomposable witnesses
- $\text{Tr}(W\rho) < 0$  = "W detects  $\rho$ "

This suggests that **if we find a fully decomposable  $W$  with  $\text{Tr}(W\rho) < 0$ , the state is GME!**

## [Example] Fully Decomposable Witnesses

Let's return to the Bell state in a two-qubit system. There is only one bipartition,  $A|B$ . A natural candidate witness for the Bell state is the **projector witness**:  $W = \frac{1}{2}I - |\Phi^+\rangle\langle\Phi^+|$ .

$$W = \frac{1}{2}I - |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

This entanglement witness detects the Bell state:

$$\text{Tr}W|\Phi^+\rangle\langle\Phi^+| = -\frac{1}{2}$$

## [Example] Fully Decomposable Witnesses

We need to check if  $W = P + Q^{T_A}$  for some  $P, Q \geq 0$ .

We can simply choose:

$$P = 0, \quad Q = W^{T_A} = \frac{1}{2}I - (|\Phi^+\rangle\langle\Phi^+|)^{T_A} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

One can verify that  $\text{Tr}(W) = 1$ , and  $P, Q \geq 0$ , confirming  $W$  is fully decomposable.

In general, finding such decompositions is non-trivial.

# [Strategy] SDP Formulation

## Semidefinite Program for GME Detection

$$\begin{aligned} & \min_{W, P_M, Q_M} \quad \text{Tr}(W\rho) \\ & \text{subject to} \quad \text{Tr}(W) = 1 \\ & \quad \quad \quad W = P_M + Q_M^{T_M}, P_M \geq 0, Q_M \geq 0 \quad \forall M \end{aligned}$$

The free parameters are  $W$  and the operators  $P_M, Q_M$  for every bipartition  $M$ . If the minimum of  $\text{Tr}(W\rho)$  is negative, then  $\rho$  is not a PPT mixture and hence is **GME**.

This optimization takes the form of a semidefinite program (SDP). The global optimality of an SDP can be certified and the solution can be efficiently computed via interior-point methods.

## Additional Strategies

- **Fully PPT witnesses:** Set  $P_M = 0$  so that  $W^{T_M} \geq 0$  for all  $M$ .
- **Observable basis:** Let  $O = \{O_1, \dots, O_k\}$  be a set of observables. Then parameterize  $W = \sum_i \lambda_i O_i$ .



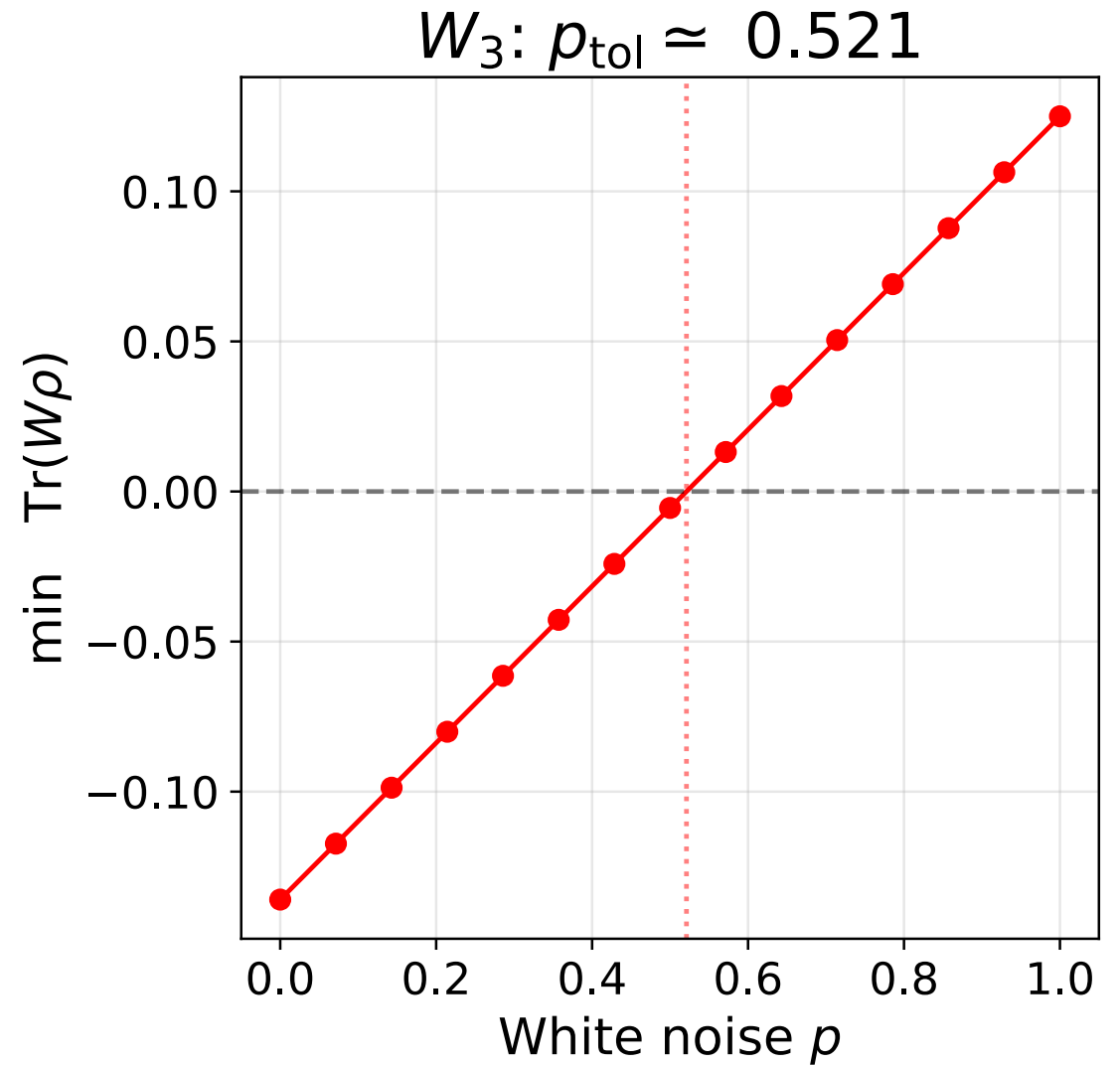
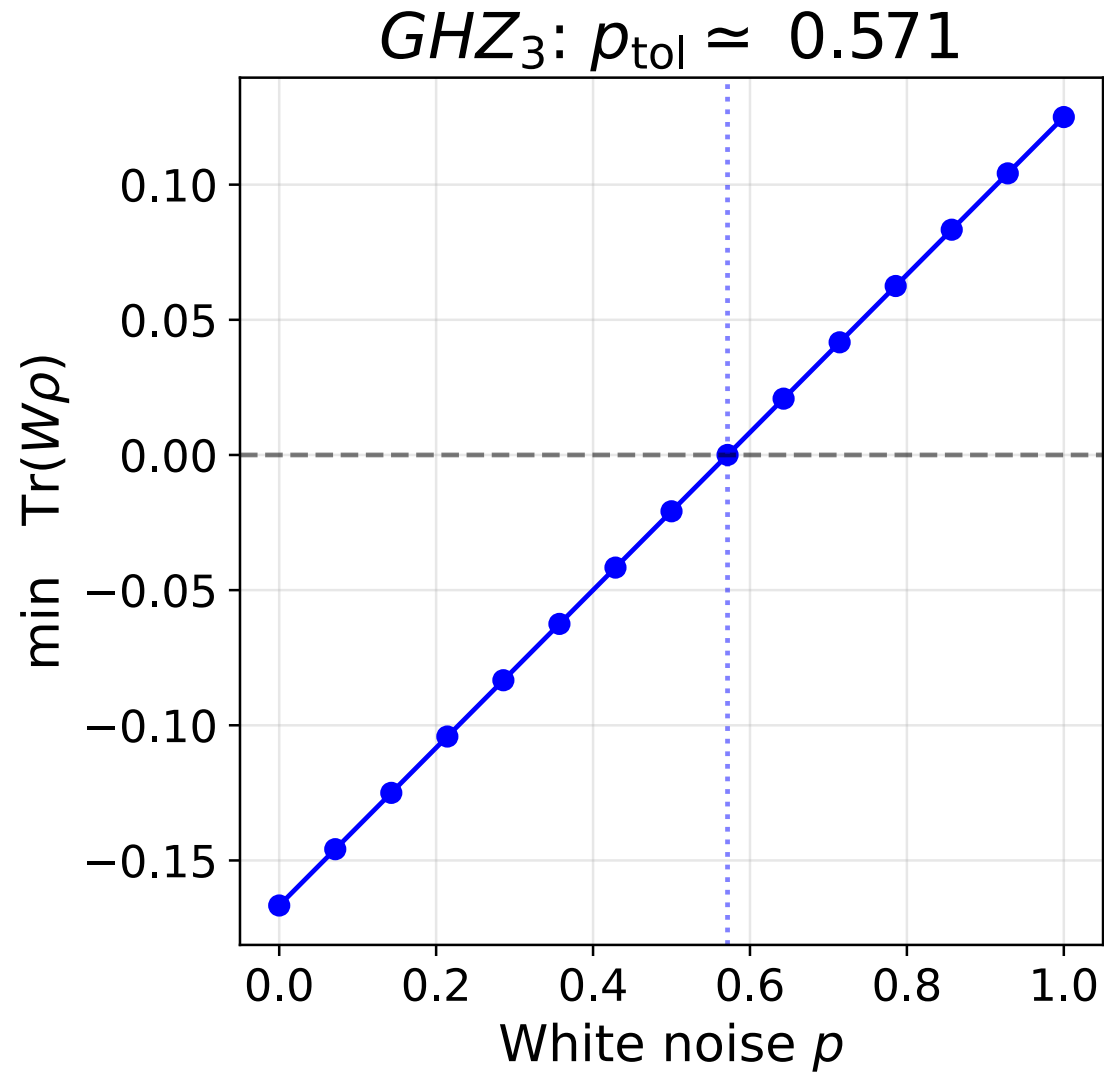
## [Results] White Noise Tolerance

Consider mixed states(isotropic state):  $\rho(p) = p \cdot \frac{I}{2^N} + (1 - p) \cdot |\psi\rangle\langle\psi|$

**Critical noise tolerance**  $p_{\text{tol}}$ : Maximum  $p$  for which GME is still detected. A higher  $p_{\text{tol}}$  indicates stronger entanglement.

| State $ \psi\rangle$ | PPT mixtures |
|----------------------|--------------|
| GHZ <sub>3</sub>     | <b>0.571</b> |
| W <sub>3</sub>       | <b>0.521</b> |
| Cluster <sub>4</sub> | <b>0.615</b> |
| Dicke <sub>4</sub>   | <b>0.539</b> |
| GHZ <sub>4</sub>     | <b>0.500</b> |

# [Numerical Simulation] White Noise Tolerance



## [Strategy] Modified SDP for Quantifying GME

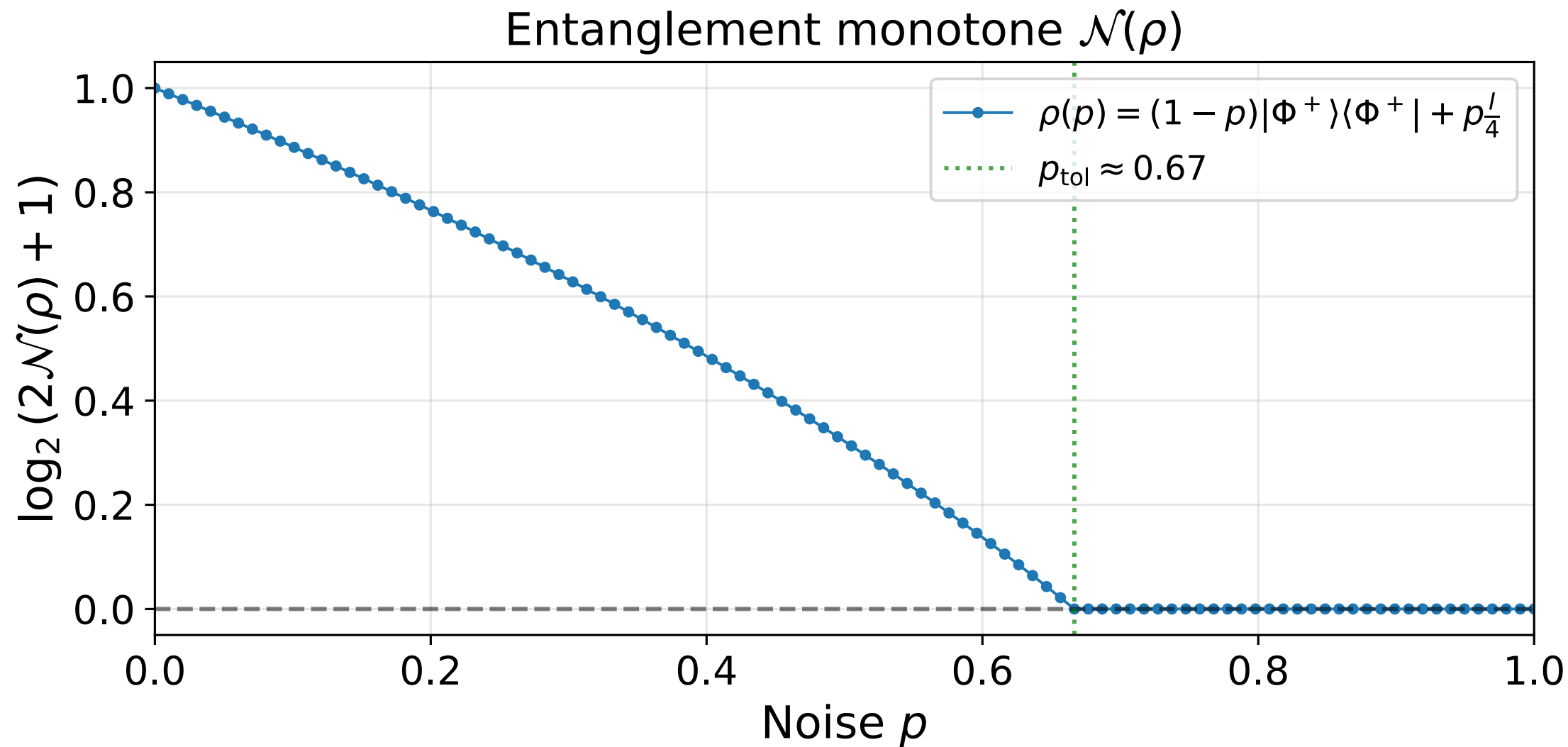
This approach can also be used to quantify GME. If the trace normalization  $\text{Tr}(W) = 1$  is replaced by  $0 \leq P_M \leq I$  and  $0 \leq Q_M \leq I$ , the negative witness expectation value becomes a multipartite entanglement monotone. Note that this quantity equals the **negativity** in the bipartite case.

$$\begin{aligned} & - \min_{W, P_M, Q_M} \text{Tr}(W\rho) \\ & \text{subject to } W = P_M + Q_M^{T_M}, 0 \leq P_M, Q_M \leq I \quad \forall M \end{aligned}$$

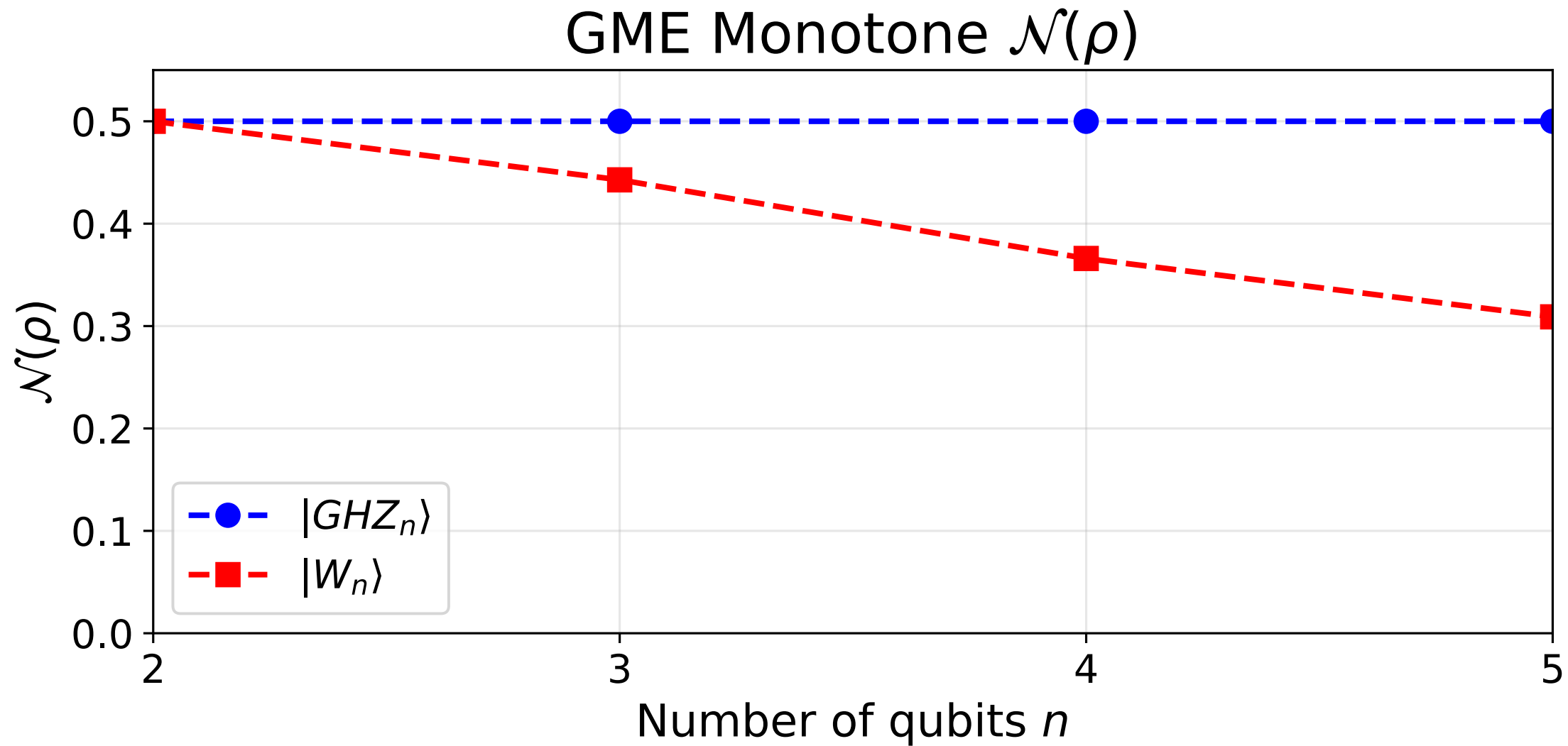
**Properties of the Monotone**  $\mathcal{N}(\rho) = - \min \text{Tr}(W\rho)$

- **Vanishes on biseparable states:**  $\mathcal{N}(\rho^{\text{bs}}) = 0$
- **Convex:**  $\mathcal{N}(\sum_i p_i \rho_i) \leq \sum_i p_i \mathcal{N}(\rho_i)$
- **LOCC monotone:** Does not increase under local operations and classical communication

# [Numerical Result] $\mathcal{N}(\rho)$ on the two-qubit isotropic state



## [Numerical Result] Quantifying GME



# Summary

## Pros

- The framework is systematic and complete.
- It provides a unified SDP-based approach for detecting GME.
- No false negatives within the PPT mixture class.
- The entanglement monotone not only detects but also quantifies GME.

# Summary

## Cons

- Generally needs full density matrix  $\rho$  — expensive quantum state tomography required
- No Closed-Form Solution. SDP optimization (convex optimization) is required for each state.
  - Time Complexity:  $\mathcal{O}(\sqrt{m}D^3(m + D^2))$
  - Space Complexity:  $\mathcal{O}(mD^2 + D^4)$

where  $D$  is the Hilbert space dimension and  $m$  is the number of constraints.

- Not Complete for All Entanglement. Only detects states outside PPT mixtures; PPT entangled states are missed

# References

## Main Paper

- Jungnitsch, Bastian, Tobias Moroder, and Otfried Gühne. "Taming multiparticle entanglement." Physical review letters 106.19 (2011): 190502.

## Related Works

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- Horodecki, Michał, Paweł Horodecki, and Ryszard Horodecki. "Separability of n-particle mixed states: necessary and sufficient conditions in terms of linear maps." Physics Letters A 283.1-2 (2001): 1-7.
- Gühne, Otfried, and Géza Tóth. "Entanglement detection." Physics Reports 474.1-6 (2009): 1-75.

## Software

- CVXPY: <https://www.cvxpy.org/>



# **Thank You!**

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Questions & Discussion